D.M. MEDICAL ONCOLOGY — PAPER IV

RECENT ADVANCES · EXAM ANSWER NOTES

Antibody drug conjugates in medical oncology practice.

February 2025 | DM 0225 | Q1 (DEEP Essay / Elaborate (FOCUS 2022–2026)) | **15 marks** | Target ~16 pages

*Includes: mechanisms · doses/regimens · landmark clinical trials · hallmarks · **NCCN 2026 / 2027 updates** · flow charts · exam pearls*

0. How to use this DEEP answer

Sections below are **high-yield, trial-dense content** written for a 15-mark Paper IV elaboration. After the deep core, additional revision modules (algorithms, supportive care, flash cards) extend the note to full booklet length.

EXAM PEARLS / HALLMARKS

- In the exam hall, prioritise the DEEP trial tables + NCCN 2026 boxes first.
- Draw at least one flowchart from the deep sections.

1. Introduction & definition (open the 15-mark answer)

Antibody–drug conjugates (ADCs) are targeted chemotherapeutics: a monoclonal antibody delivers a highly potent cytotoxic payload to antigen-expressing tumor cells, improving the therapeutic index versus conventional cytotoxics. They now span solid tumors (breast, lung, gastric, urothelial, ovarian) and hematologic malignancies (Hodgkin/T-cell lymphoma, DLBCL, myeloma).

For Paper IV, structure the answer as: (i) ADC anatomy & bystander biology → (ii) catalogue by antigen/payload → (iii) landmark phase III trials with doses → (iv) class toxicities (ILD, neuropathy, ocular, cytopenias) → (v) NCCN 2026 expansions → (vi) sequencing pearls.

- Definition one-liner: mAb + linker + payload ± bystander kill of antigen-low neighbors.
- Exam currency phrase: ‘ADC footprint expanded across NCCN Guidelines® 2026/2027.’
- Contrast briefly with naked mAbs, BiTEs, and CAR-T (delivery of toxin vs immune engagement).

EXAM PEARLS / HALLMARKS

- Draw the three-part ADC cartoon in the first 2 minutes — examiners reward mechanism diagrams.
- Always pair each named ADC with one trial + one dose + one signature toxicity.

2. ADC structure — antibody, linker, payload, DAR

Antibody: usually IgG1 recognizing a surface antigen enriched on tumor (HER2, TROP2, Nectin-4, folate receptor-α, CD30, CD79b, BCMA/BCMA-adjacent, CD19). Internalization after binding is required for most payloads; antigen density, trafficking, and shedding modulate efficacy.

Linker: cleavable (enzyme-/pH-/reducible-sensitive) vs non-cleavable. Cleavable linkers liberate membrane-permeable toxins → bystander effect. Non-cleavable linkers (e.g., T-DM1 thioether) release active catabolites mainly inside the lysosome → less bystander, more antigen-restricted kill.

Payload: microtubule inhibitors (DM1, MMAE, MMAF) or topoisomerase-I inhibitors (DXd, SN-38). Drug-to-antibody ratio (DAR) typically 2–8; higher DAR and permeable topo-I payloads partly explain activity in HER2-low disease with T-DXd.

ADC cellular itinerary

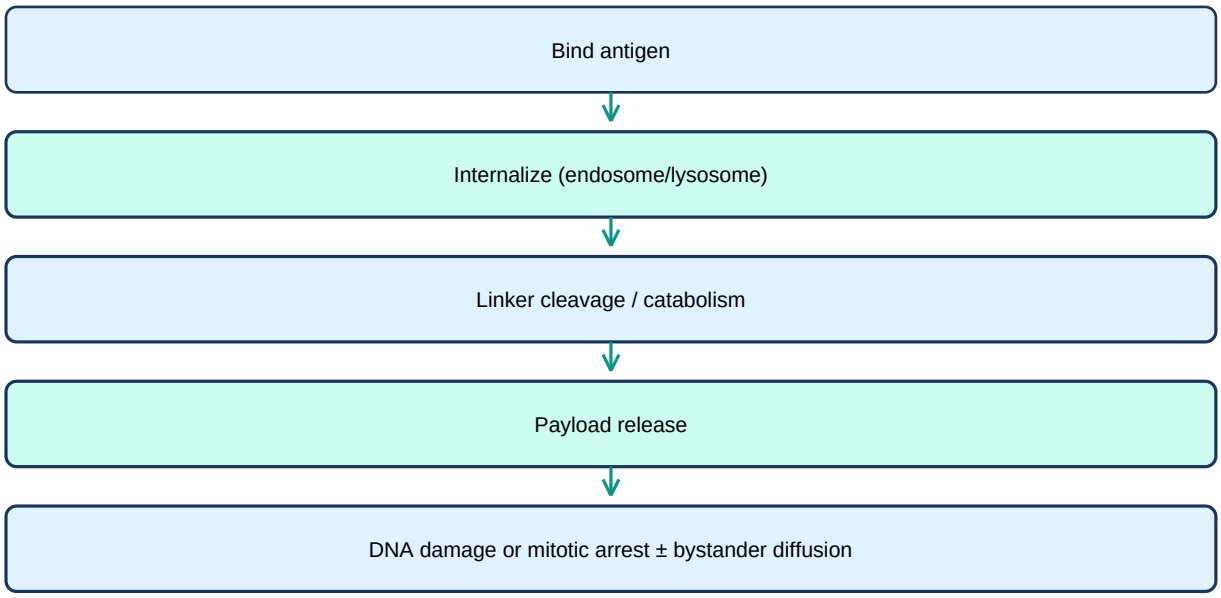


Figure: stepwise exam algorithm

mAb	Targets tumor antigen; Fc may add ADCC
Linker	Cleavable → bystander; non-cleavable → intracellular
Payload	Microtubule toxin or topo-I poison
DAR	Payload molecules per Ab (design trade-offs)
Internalization	Lysosomal processing → free toxin

Figure: mechanism / pathway map

3. Bystander effect & HER2-low paradigm

Bystander killing: membrane-permeable payloads (DXd, SN-38, some MMAE contexts) diffuse to neighboring antigen-negative/low cells within heterogeneous tumors. This underpins DESTINY-Breast04/06 activity in HER2-low (IHC 1+ or 2+/ISH–) and ultralow disease — a conceptual leap beyond classical HER2-positive thresholds.

Resistance themes: antigen downregulation/heterogeneity, impaired internalization, payload efflux (MDR1), lysosomal changes, and payload-pathway alterations. Sequencing across ADCs with non-overlapping payloads (microtubule → topo-I or vice versa) is an active clinical question.

- T-DM1: limited bystander → needs stronger HER2 expression historically.
- T-DXd / SG / Dato-DXd: cleavable + permeable payload → broader antigen spectrum.
- Report HER2-low explicitly on pathology — changes metastatic options.

EXAM PEARLS / HALLMARKS

- One exam sentence: ‘Bystander effect + high DAR explain T-DXd benefit beyond IHC 3+/ISH+.’

4. HER2 ADCs — T-DXd and T-DM1

Ado-trastuzumab emtansine (T-DM1): trastuzumab + non-cleavable linker + DM1. Landmark EMILIA (post-trastuzumab metastatic) and KATHERINE (adjuvant residual disease after neoadjuvant HER2 therapy). Dose 3.6 mg/kg IV Q3W. Toxicities: thrombocytopenia, transaminitis, cardiotoxicity monitoring continues.

Trastuzumab deruxtecan (T-DXd): cleavable linker + DXd (topo-I), DAR ≈8, potent bystander. DESTINY-Breast03 established superiority over T-DM1 in 2L HER2+ mBC (mPFS ~28.8 vs 6.8 mo). DESTINY-Breast04/06 expanded to HER2-low/ultralow. DESTINY-Gastric01/04 support gastric/GEJ after trastuzumab (often 6.4 mg/kg). DESTINY-Lung / NSCLC HER2-mutated niches; urothelial HER2-expressing subsequent lines evolving in NCCN.

Breast metastatic dose typically 5.4 mg/kg IV Q3W. Mandatory ILD/pneumonitis counseling and CT monitoring.

Trial	Key result / exam takeaway
EMILIA	T-DM1 > lapatinib+capecitabine post-trastuzumab
KATHERINE	Adjuvant T-DM1 > trastuzumab if residual disease
DESTINY-Breast03	T-DXd ■ T-DM1 2L HER2+ mBC
DESTINY-Breast04	T-DXd practice-changing in HER2-low
DESTINY-Breast06	Earlier HR+/HER2-low–ultralow, chemo-naive metastatic
DESTINY-Gastric04	OS benefit T-DXd 2L HER2+ gastric after trastuzumab

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
T-DM1	3.6 mg/kg IV Q3W
T-DXd (breast)	5.4 mg/kg IV Q3W — hold/discontinue per ILD grade
T-DXd (gastric)	6.4 mg/kg IV Q3W per label/indication
Trastuzumab (context)	8 → 6 mg/kg IV Q3W with pertuzumab CLEOPATRA backbone

Table: doses and regimens

5. TROP2 ADCs — sacituzumab govitecan & datopotamab deruxtecan

Sacituzumab govitecan (SG): anti-TROP2 + hydrolyzable linker + SN-38 (irinotecan active metabolite). ASCENT: SG vs physician's-choice chemo in relapsed TNBC — PFS/OS benefit. TROPiCS-02: HR+/HER2– metastatic breast after endocrine and chemo. Also approved/used in pretreated urothelial carcinoma. Dose 10 mg/kg IV D1,8 Q21D.

Datopotamab deruxtecan (Dato-DXd): TROP2-directed DXd ADC. TROPION-Lung program and NCCN 2026 NSCLC pathway place Dato-DXd as preferred subsequent therapy after osimertinib + chemotherapy in EGFR-mutated disease, and as preferred options in later lines — high-yield 2026 currency point.

- SG pearls: early diarrhea/neutropenia management; UGT1A1 caution analogous to irinotecan biology.
- Do not confuse SG (SN-38) with DXd ADCs — shared TROP2 target, different payloads/toxicity profiles.

Trial	Key result / exam takeaway
ASCENT	SG > chemo in R/R TNBC — PFS/OS
TROPiCS-02	SG in HR+/HER2– mBC post-endocrine/chemo
TROPION-Lung datasets	Dato-DXd activity → NCCN preferred post-osi+chemo

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Sacituzumab govitecan	10 mg/kg IV D1,8 every 21 days
Dato-DXd	Weight-based IV Q3W per trial/label — monitor ILD/stomatitis

Table: doses and regimens

6. Nectin-4 & FRα — enfortumab vedotin & mirvetuximab

Enfortumab vedotin (EV): anti-Nectin-4 + MMAE. EV-301 established OS in previously treated urothelial cancer. EV-302 (EV + pembrolizumab) remade 1L la/mUC — now sole preferred metastatic regimen in NCCN Bladder 2026. Perioperative EV+P data (KEYNOTE-905/EV-303) for cisplatin-ineligible MIBC. Dose 1.25 mg/kg IV

D1,8,15 Q28D (adjust with pembrolizumab schedules).

Mirvetuximab soravtansine: anti–folate receptor-α + maytansinoid. MIRASOL: OS benefit in FRα-high platinum-resistant ovarian cancer vs chemo. Ocular toxicity (keratopathy) requires proactive ophthalmologic care and dose holds. NCCN ovarian pathways incorporate mirvetuximab in FRα+ platinum-resistant/sensitive contexts alongside 2026 updates (ROSELLA, KEYNOTE-B96).

EV / mirvetuximab clinical hooks

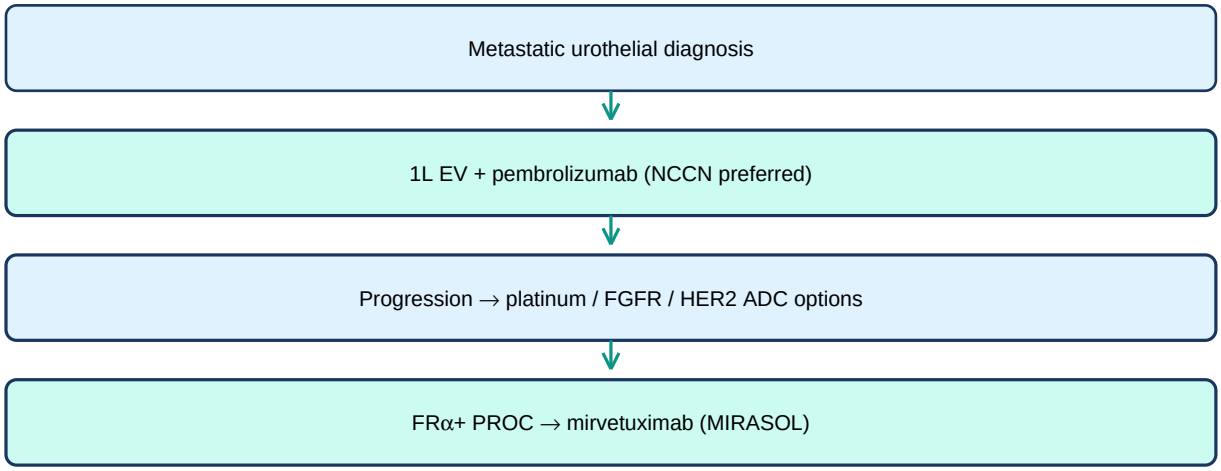


Figure: stepwise exam algorithm

Trial	Key result / exam takeaway
EV-301	EV > chemo previously treated urothelial — OS
EV-302	EV + pembrolizumab 1L la/mUC — new standard
KEYNOTE-905 / EV-303	Perioperative EV+P cisplatin-ineligible MIBC
MIRASOL	Mirvetuximab > chemo FRα-high PROC — OS

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Enfortumab vedotin	1.25 mg/kg IV D1,8,15 Q28D (± pembrolizumab)
Mirvetuximab	6 mg/kg adjusted ideal body weight IV Q3W — ocular monitoring

Table: doses and regimens

7. Hematologic ADCs — brentuximab, polatuzumab, belantamab, loncastuximab

Brentuximab vedotin (BV): anti-CD30 + MMAE. ECHELON-1 (BV-AVD in advanced Hodgkin), ECHELON-2 (BV-CHP in CD30+ PTCL), and relapsed HL/ALCL indications. Neuropathy is dose-limiting; PML rare but boxed historically with related agents.

Polatuzumab vedotin: anti-CD79b + MMAE. POLARIX: pola-R-CHP vs R-CHOP in previously untreated DLBCL — PFS benefit; integrate per risk/regulatory context. Also used in transplant-ineligible R/R DLBCL combinations.

Belantamab mafodotin: anti-BCMA + MMAF (non-cleavable). Ocular keratopathy hallmark. DREAMM-7 (belantamab + Vd) informs NCCN Myeloma 2026/2027 Cat 1 other recommended after ≥2 prior lines (PI+IMiD).

Loncastuximab tesirine: anti-CD19 + PBD dimer payload for R/R large B-cell lymphoma after ≥2 lines (LOTIS program). Edema/effusions and myelosuppression need vigilance; PBD class differs from MMAE neuropathy pattern.

Trial	Key result / exam takeaway
ECHELON-1 / ECHELON-2	BV in 1L HL / CD30+ PTCL

POLARIX	Pola-R-CHP vs R-CHOP 1L DLBCL
DREAMM-7	Belantamab + Vd — NCCN myeloma listing
LOTIS-2	Loncastuximab in R/R LBCL

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Brentuximab vedotin	1.2 mg/kg with AVD (ECHELON-1) or 1.8 mg/kg Q3W R/R schedules
Polatuzumab vedotin	1.8 mg/kg IV Q3W in pola-R-CHP / R/R combos
Belantamab mafodotin	Combination dosing per DREAMM-7 — ocular REMS-style monitoring
Loncastuximab tesirine	0.15 mg/kg Q3W ×2 then 0.075 mg/kg Q3W (label schedule)

Table: doses and regimens

8. Cross-tumor ADC catalogue (exam table to redraw)

Present a single comparison table the examiner can tick. Columns: ADC | Target | Payload | Key trial | Signature toxicity | NCCN niche.

- T-DXd — HER2 — DXd — DESTINY program — ILD — breast/gastric/NSCLC/urothelial niches
- T-DM1 — HER2 — DM1 — EMILIA/KATHERINE — platelets/LFTs — residual HER2+ EBC; selected metastatic
- SG — TROP2 — SN-38 — ASCENT/TROPiCS — neutropenia/diarrhea — TNBC & HR+ mBC; urothelial
- Dato-DXd — TROP2 — DXd — TROPION — ILD/stomatitis — EGFR+ NSCLC subsequent (NCCN 2026)
- EV — Nectin-4 — MMAE — EV-302 — neuropathy/rash/hyperglycemia — 1L urothelial ± peri-op
- Mirvetuximab — FRα — maytansinoid — MIRASOL — ocular — FRα+ ovarian
- BV — CD30 — MMAE — ECHELON — neuropathy — HL/PTCL
- Pola — CD79b — MMAE — POLARIX — neuropathy/cytopenias — DLBCL
- Belantamab — BCMA — MMAF — DREAMM-7 — ocular — myeloma combinations
- Loncastuximab — CD19 — PBD — LOTIS — edema/myelosuppression — R/R LBCL

EXAM PEARLS / HALLMARKS

- Microtubule payloads → neuropathy; DXd → ILD; MMAF/mirvetuximab → ocular; SN-38 → diarrhea/neutropenia.

9. Toxicities — ILD, neuropathy, ocular, metabolic

ILD/pneumonitis: class effect of deruxtecan ADCs (T-DXd, Dato-DXd). Grade-based algorithm: hold for suspected ILD, start steroids promptly for ≥grade 2, permanently discontinue for grade 3–4 per labels. Educate patients on cough/dyspnea/fever; low threshold for HRCT.

Peripheral neuropathy: MMAE ADCs (EV, BV, polatuzumab) and taxane carryover. Dose reduce/delay; distinguish from CPI neurologic irAEs when combinations used (EV+P).

Ocular: mirvetuximab and belantamab — lubricating drops, dose holds, ophthalmology partnerships. Hyperglycemia and skin toxicity notable with EV. Neutropenia/diarrhea with SG — growth factors and early loperamide protocols.

- Cardiac monitoring still relevant when HER2 ADCs follow anthracyclines.
- Infusion reactions: premedication per product monograph.
- Infection risk with cytopenias — especially hematologic ADCs + steroids for ILD.

ADC toxicity management loop

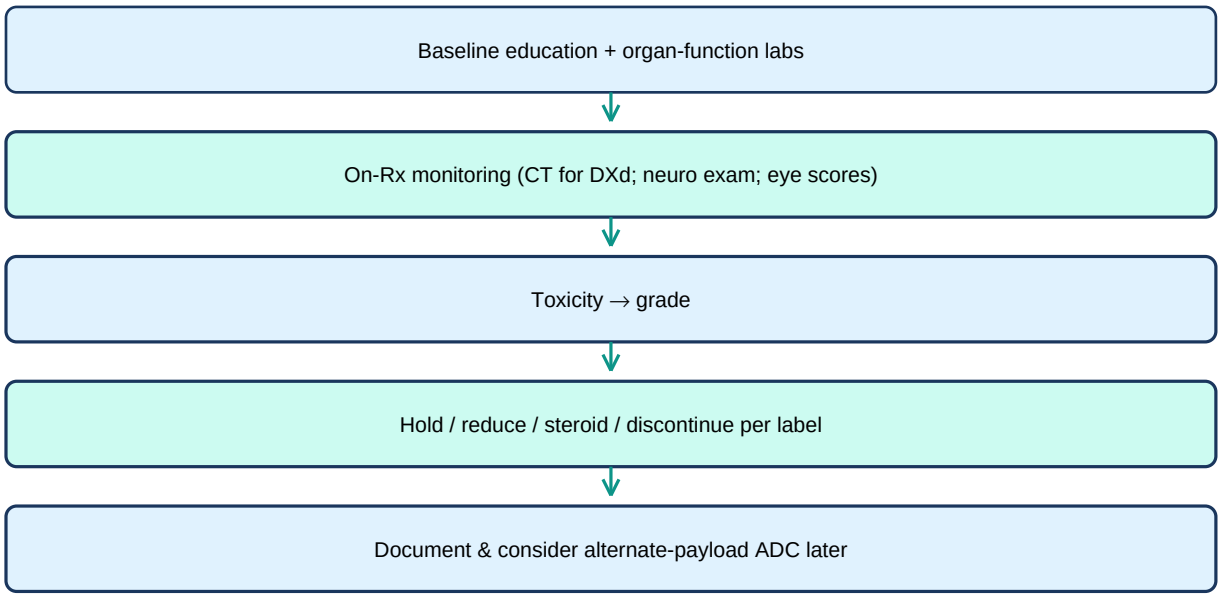


Figure: stepwise exam algorithm

10. Sequencing principles across diseases

Breast HER2+: dual blockade 1L (CLEOPATRA) → T-DXd 2L (DB-03) preferred over T-DM1; tucatinib triplet for CNS-predominant progression (HER2CLIMB). Adjuvant residual disease still T-DM1 (KATHERINE) — different disease setting.

TNBC / HR+: SG after prior lines; T-DXd if HER2-low; pembrolizumab combinations when CPS eligible; PARPi if gBRCA — avoid stacking overlapping marrow toxicity without plan.

Urothelial: EV+P first; genomic switchboard after progression (FGFR → erdafitinib; HER2 → T-DXd). Ovarian: FRα → mirvetuximab; HER2-expressing → T-DXd discussion; platinum-resistant non-ADC options include nab-paclitaxel+relacorilant (ROSELLA) per NCCN 2026.

ADC sequencing skeleton

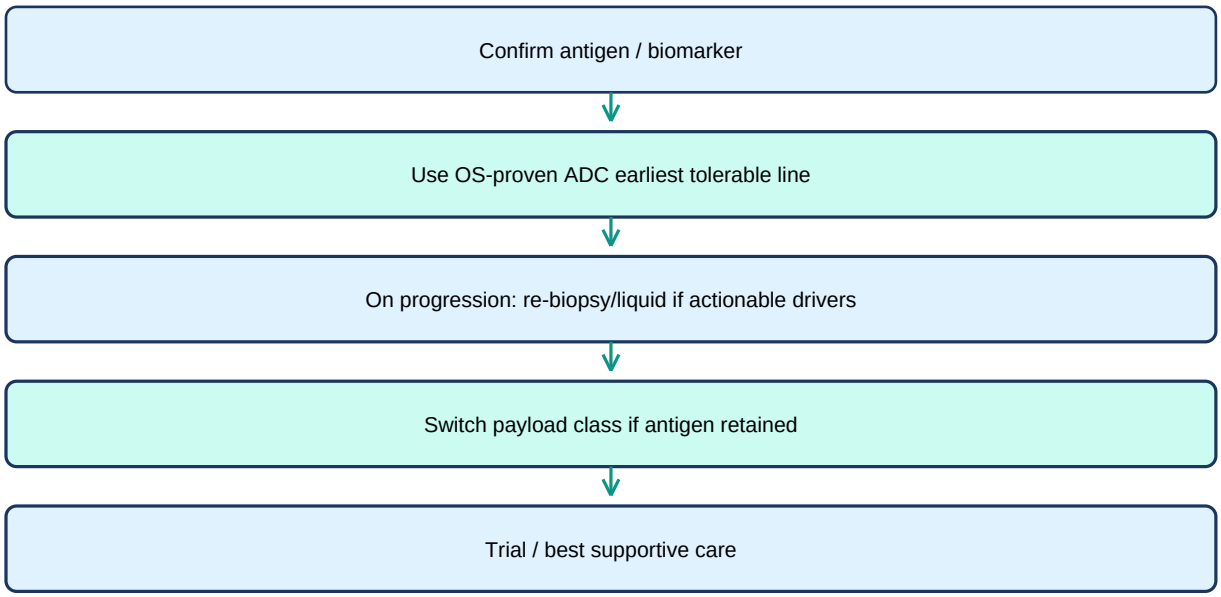


Figure: stepwise exam algorithm

11. Landmark trials master table

Memorize five ‘must-quote’ programs: DESTINY (T-DXd), ASCENT & TROPiCS (SG), EV-302 (EV+P), MIRASOL (mirvetuximab). Add KATHERINE/EMILIA for T-DM1 and POLARIX/ECHELON/DREAMM/LOTIS for heme.

Trial	Key result / exam takeaway
DESTINY-Breast03/04/06	T-DXd reshapes HER2+ and HER2-low breast
ASCENT	SG OS benefit R/R TNBC
TROPiCS-02	SG in HR+/HER2– mBC
EV-302	EV+P sole preferred 1L metastatic UC (NCCN)
MIRASOL	Mirvetuximab OS in FRα-high PROC
KATHERINE	Adjuvant T-DM1 residual HER2+ EBC
POLARIX	Pola-R-CHP 1L DLBCL
DREAMM-7	Belantamab+Vd myeloma — NCCN 2026/27

Table: landmark clinical trials

12. NCCN 2026 / 2027 — ADC expansions (boxed content)

State versions explicitly. ADC-related bullets appear across Breast v9.2026, NSCLC v9.2026, Gastric v4.2026, Bladder v3.2026, Ovarian 2026 updates, Myeloma v2.2027, and B-cell lymphoma pathways.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Breast Cancer v9.2026; NSCLC v9.2026; Gastric v4.2026; Bladder Cancer v3.2026; Ovarian Cancer 2026 panel updates; Multiple Myeloma v2.2027; B-Cell Lymphomas pathways
- T-DXd pathways: HER2+ and HER2-low/ultralow breast; HER2+ gastric 2L; HER2-mut/expressing NSCLC & urothelial subsequent niches.
- Dato-DXd preferred after osimertinib + chemo in EGFR+ NSCLC (and listed across EGFR alteration subsequent settings).
- SG retained in TNBC/HR+ breast and urothelial algorithms.
- EV + pembrolizumab sole preferred 1L la/mUC; perioperative EV+P for cisplatin-ineligible MIBC.
- Mirvetuximab for FRα+ ovarian; belantamab combinations in myeloma; loncastuximab in R/R LBCL; polatuzumab in DLBCL.
- Cross-cutting: expanding ADC footprint is an explicit NCCN 2026 theme for exam answers.

EXAM PEARLS / HALLMARKS

- Write ‘NCCN 2026’ next to EV-302 and Dato-DXd — signals currency for Paper IV.

13. Special situations & Indian/NCG practical notes

Access and cost: ADCs are expensive; discuss biosimilar trastuzumab backbones before ADC lines, trial enrollment, and center capability for ILD steroid protocols. CDSCO approvals may lag FDA/EMA — reconcile in the answer.

Pregnancy: cytotoxic payloads contraindicated. Breastfeeding contraindicated. Fertility counseling before MMAE/topo-I ADCs. Elderly: neuropathy risk with EV/BV; ILD risk with DXd in smokers/prior chest RT — individualize.

- MDT + ophthalmology/pulmonology co-management pathways.
- Document antigen assay method (IHC clone, FRα scoring, HER2-low).
- Pharmacovigilance: report ILD and ocular events.

14. Summary checklist & exam pearls

Close with a numbered summary: structure → bystander → 10 named ADCs → 5 trial acronyms → doses → toxicities → NCCN 2026.

- Anatomy: Ab + linker + payload ± bystander
- Solid tumor stars: T-DXd, T-DM1, SG, Dato-DXd, EV, mirvetuximab
- Heme stars: brentuximab, polatuzumab, belantamab, loncastuximab
- Trials: DESTINY, ASCENT, TROPiCS, EV-302, MIRASOL
- Tox: ILD (DXd), neuropathy (MMAE), ocular (mirv/belantamab)
- NCCN 2026: Dato-DXd EGFR NSCLC; EV+P bladder; ADC catalogue expansion

15-mark ADC answer skeleton

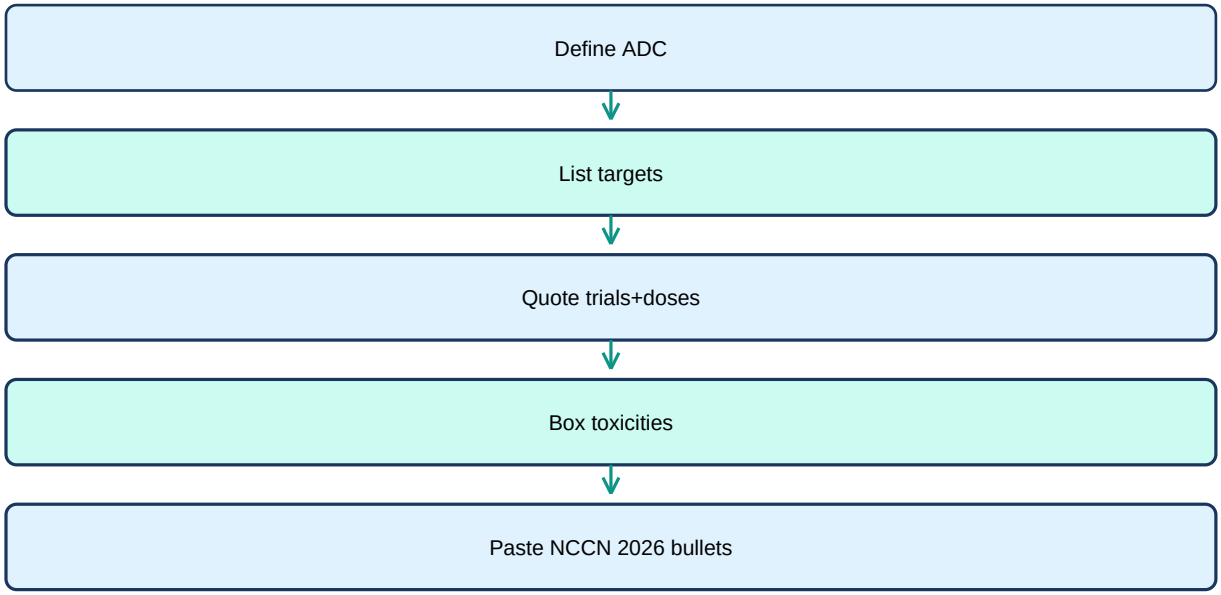


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- KATHERINE ≠ metastatic sequencing — adjuvant residual disease only.
- EV neuropathy + pembrolizumab irAE overlap needs careful grading.
- HER2-low is an ADC story, not classical trastuzumab monotherapy story.
- Payload class switch may salvage antigen-persistent progression.
- Always mention monitoring: HRCT/steroids for ILD; eye exams for FRα/BCMA ADCs.

— Additional exam-length revision modules —

The following modules reinforce algorithms, toxicities, Indian/NCG context, and last-page summary structure.

3. Diagnosis, biomarkers & staging

A modern oncology answer must separate **diagnostic**, **prognostic**, and **predictive** markers. Predictive markers change drug choice (EGFR, HER2, BRCA, MSI, Claudin18.2, FGFR, NTRK).

- Tissue diagnosis with adequate IHC panel; never treat metastatic disease without molecular profiling when actionable targets exist.
- Use validated assays (FDA/companion diagnostics where mandated).
- Stage with AJCC 8th/9th as applicable (NSCLC NCCN 2026 uses AJCC 9th).

• Performance status, organ function, comorbidities, geriatric assessment, and goals of care before regimen selection.

Work-up algorithm (draw this in the answer booklet)

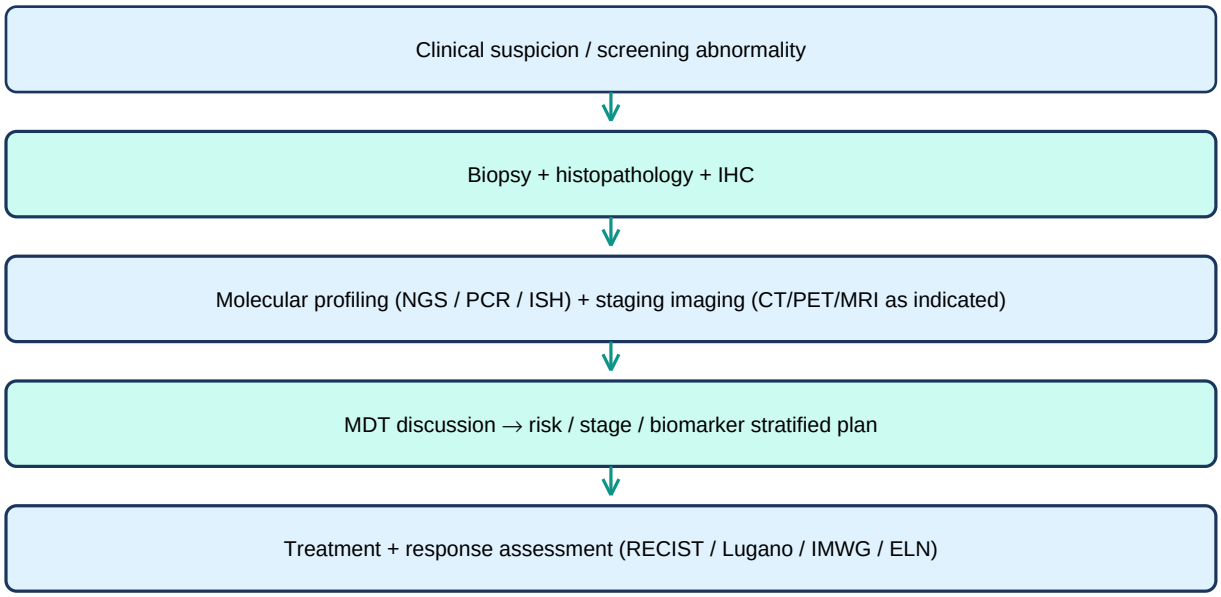


Figure: stepwise exam algorithm

4. Management algorithm, doses & regimens

Structure management as: (A) localized/curative-intent pathways, (B) advanced/metastatic systemic therapy, (C) special populations (CNS, elderly, organ dysfunction, pregnancy), (D) supportive care.

For each line of therapy state: eligibility, preferred regimen, dose, key toxicity monitoring, and the trial that supports it.

• Localized disease: multimodality — surgery/RT/systemic as per disease chapter.

• Neoadjuvant vs adjuvant: cite pathologic response and adjuvant switch trials (e.g., KATHERINE, CheckMate 577, KEYNOTE-522).

• Metastatic: biomarker-first; avoid empiric chemo when a high-level targeted/IO option exists.

• Oligometastatic disease: consider metastasis-directed therapy where NCCN supports (e.g., prostate APMS 2026).

• Re-biopsy / liquid biopsy at progression to map resistance.

Management flowchart (reproduce in exam)

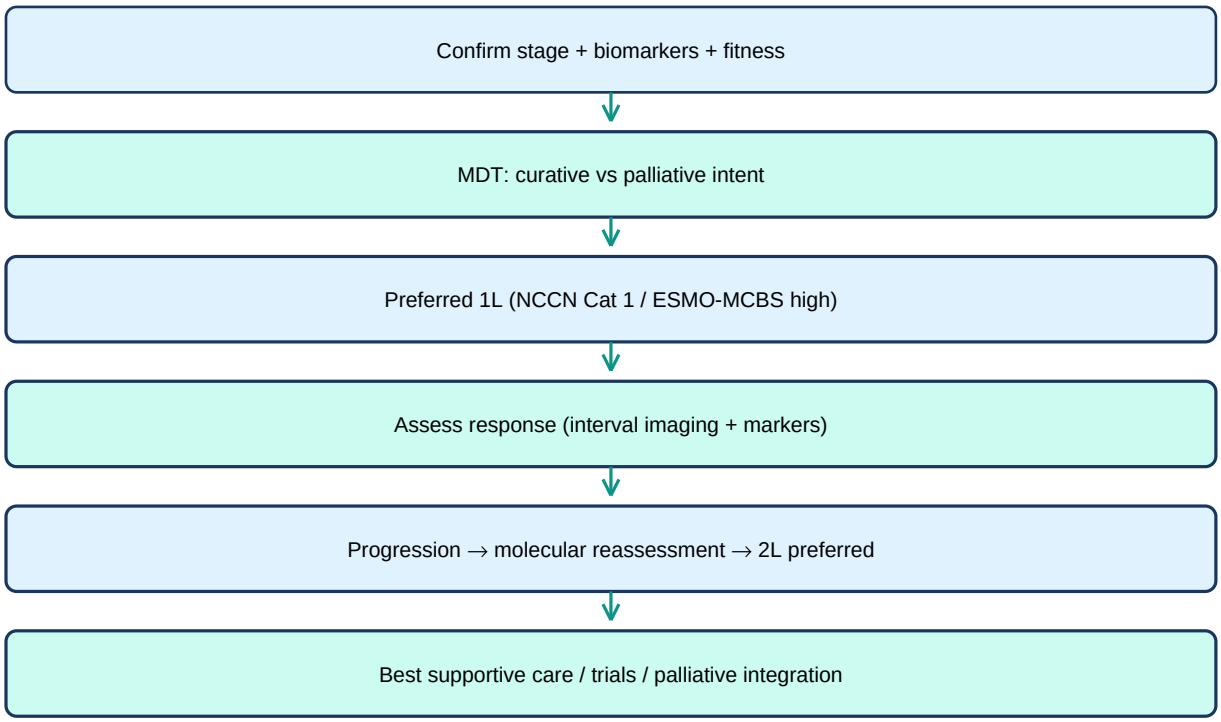


Figure: stepwise exam algorithm

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

5. Landmark clinical trials (must quote in answer)

Present trials in a table: name → design/population → intervention → primary endpoint → absolute/relative benefit → practice change. Examiners award marks for correct trial names and one numeric result (HR or median PFS/OS).

- Mention phase (III preferred for standards) and whether endpoint was PFS, OS, MRD, pCR, or EFS.
- Distinguish statistical vs clinical significance; note crossover and subgroup caveats.
- For India-focused answers, mention access/biosimilars/NCG guidance after the trial table.

Trial	Key result / exam takeaway
HERA / BCIRG-006 / NSABP B-31 / N9831	Adjuvant trastuzumab OS/DFS benefit
CLEOPATRA	Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MOSAIC / C-07 / IDEA	Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX
ATOMIC	FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred
ZUMA-1 / JULIET / TRANSCEND	Axi-cel; tisa-cel; liso-cel LBCL
ZUMA-7 / TRANSFORM / BELINDA	2L CAR-T vs SOC

Table: landmark clinical trials

6. Toxicities, supportive care & monitoring

A complete answer links efficacy to safety. List monitoring schedules (CBC, chemistry, ECG, TFTs, cortisol, LFTs, PFTs/CT for ILD risk drugs).

- Anticipate class toxicities and counsel patients; use CTCAE grading.
- Cardio-oncology: anthracyclines, HER2 mAbs, ibrutinib, checkpoint myocarditis — baseline ECG/echo when indicated.
- IO toxicities: irAEs — hold drug, steroids, specialty referral; endocrine irAEs often permanent replacement.
- ADC pulmonary toxicity (T-DXd ILD): hold permanently for ≥grade 2 in many protocols; imaging + steroids.
- TLS prophylaxis for venetoclax/BLIN/CAR-T; rasburicase if high-risk.
- Fertility, sexuality, survivorship, and psychosocial support are part of comprehensive answers.

Toxicity management loop

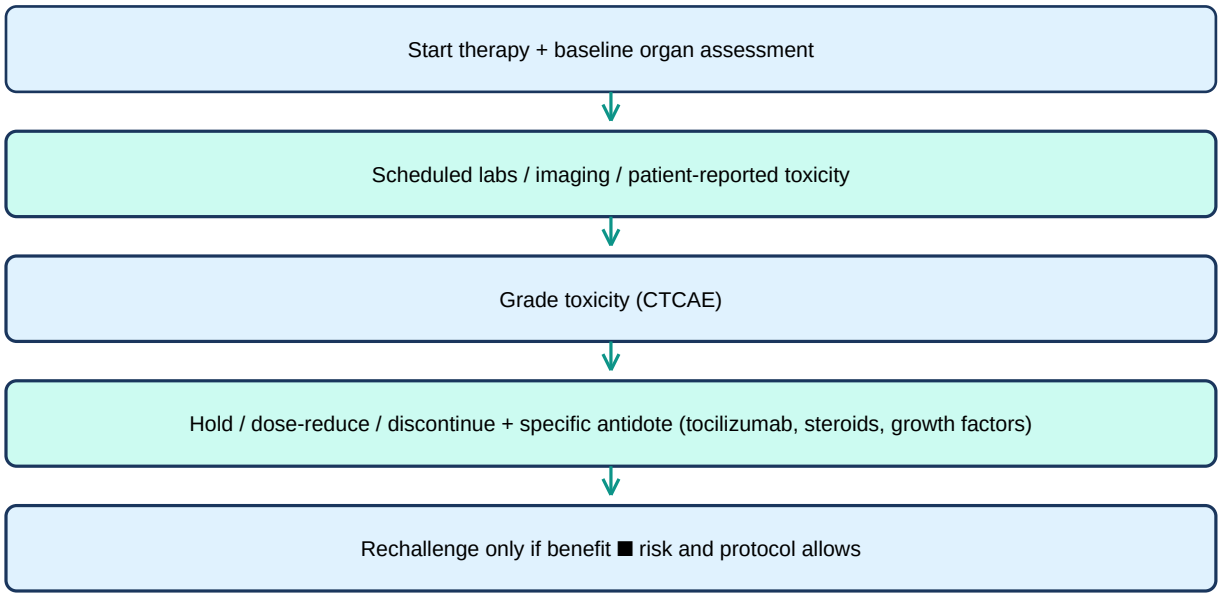


Figure: stepwise exam algorithm

7. NCCN 2026 / 2027 updates (high-yield for ‘recent advances’)

Paper IV rewards currency. Explicitly write: **“As per NCCN Guidelines updates in 2026/2027…”** and list 3–5 bullets. Also mention ESMO guidelines / Indian NCG where relevant for balance.

- State Category (1 / 2A / 2B) and preferred vs other recommended vs useful in certain circumstances.
- If Indian exam: reconcile NCCN with CDSCO approvals and NCG practical pathways.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:*
- Expanding ADC footprint across NCCN 2026: T-DXd (HER2 breast/gastric/NSCLC/urothelial), Dato-DXd (EGFR+ NSCLC), sacituzumab govitecan (breast/urothelial), mirvetuximab (ovarian FRα), enfortumab vedotin (urothelial), belantamab combinations (myeloma), loncastuximab (B-cell).

EXAM PEARLS / HALLMARKS

- NCCN versions change frequently — cite year/version (e.g., NSCLC v9.2026, Breast v9.2026, Myeloma v2.2027).
- Category 1 = high-level evidence + uniform NCCN consensus.

8. Special situations, differentials & practical issues

Discuss differentials, mimics, and when to refer to radiation oncology, surgery, palliative care, genetics, or cell-therapy centers.

Include brief notes on health economics (cost-effectiveness, biosimilars) and clinical trial enrollment as standard options.

- MDT documentation improves outcomes and exam completeness.
- Survivorship care plans: late effects, second malignancies, cardiac, endocrine, psychosocial.
- Real-world evidence and digital tools increasingly complement RCT data (relevant to 2026 Paper IV themes).

9. Evidence hierarchy, patient rights & benefits

Quote evidence using: Phase III RCT > meta-analysis of RCTs > prospective registries / RWE > retrospective > expert consensus. NCCN Categories and ESMO-MCBS scores help communicate strength of benefit.

Patient-centered care: informed consent for toxicities (especially CRS/ICANS, ILD, infertility), shared decision-making, financial toxicity counseling, and access to supportive/palliative care concurrent with active therapy.

- Benefits to highlight: OS, PFS, pCR, MRD negativity, organ preservation, QoL, reduced hospitalization.
- Rights: autonomy, confidentiality (EMR), right to second opinion, trial participation, refusal of care, end-of-life preferences.
- Equity: molecular testing access, biosimilars, NCG/Ayushman Bharat practical pathways for Indian exams.
- Pharmacovigilance: report ADRs; Phase IV / post-marketing commitments.

EXAM PEARLS / HALLMARKS

- Write one sentence on absolute benefit (e.g., ‘median OS improved from X to Y months’).)
- Mention number-needed-to-treat conceptually if space permits.

10. Summary checklist (write this last page in the booklet)

Close the 15-mark answer with a crisp numbered summary the examiner can tick.

- Definition + epidemiology one-liner
- Key biology / hallmark diagram
- Essential diagnostics & biomarkers
- Risk/stage stratification
- Preferred treatment algorithm with ≥2 doses
- ≥3 named trials with endpoints
- Major toxicities + monitoring
- NCCN 2026 update bullets (3–5)
- Special situations
- Five exam pearls

30-second revision flowchart

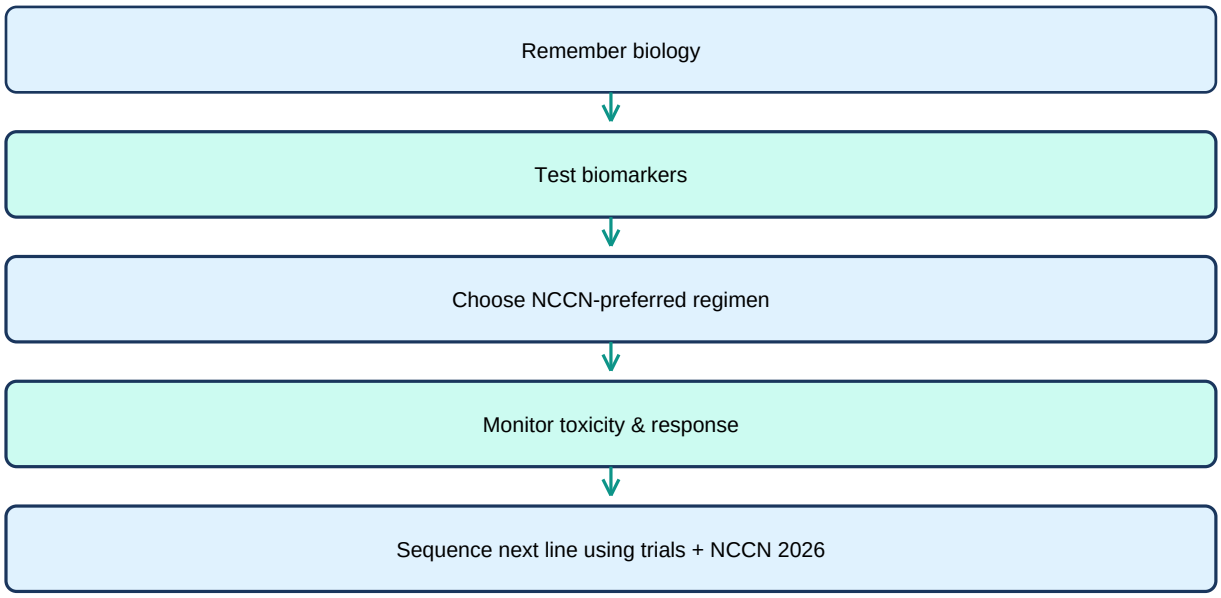


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- Neat flowchart + trial table often separates a 10/15 from a 13–15/15.
- Do not dump drug lists without doses or evidence.
- Currency: say 'NCCN 2026' explicitly for Paper IV.

11. Pharmacology deep-dive (class, PK/PD, interactions)

For drug-centered questions, present: chemical/biologic class → target epitope/kinase → PK (oral bioavailability, half-life, hepatic/renal clearance) → food effects → CYP interactions → dose adjustments → therapeutic drug monitoring if any.

Example interaction pearls: acid-suppressive therapy with some TKIs; QT-prolonging combinations; azole antifungals with vinca/TKIs; strong CYP3A inducers reducing exposure to many oral targeted agents.

- State renal/hepatic dose modifications when known.
- Note formulation differences (olaparib tablets vs capsules; IV vs SC daratumumab).
- Infusion reactions: premedication protocols for mAbs/BiTEs.

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

12. Response assessment & follow-up schedules

Use disease-specific criteria: RECIST 1.1 solid tumors; iRECIST for immunotherapy; Lugano/Deauville for lymphoma; IMWG for myeloma; ELN molecular response for CML/AML; PSA working group for prostate.

- Typical imaging interval q2–3 cycles in metastatic disease unless clinically indicated sooner.
- PET-adapted strategies in Hodgkin lymphoma (RATHL, AHL2011) — cite Deauville 1–3 negative vs 4–5 positive.

• ctDNA MRD emerging for escalation/de-escalation (CRC, lymphoma, breast — exam discussion point).

Clinical algorithm

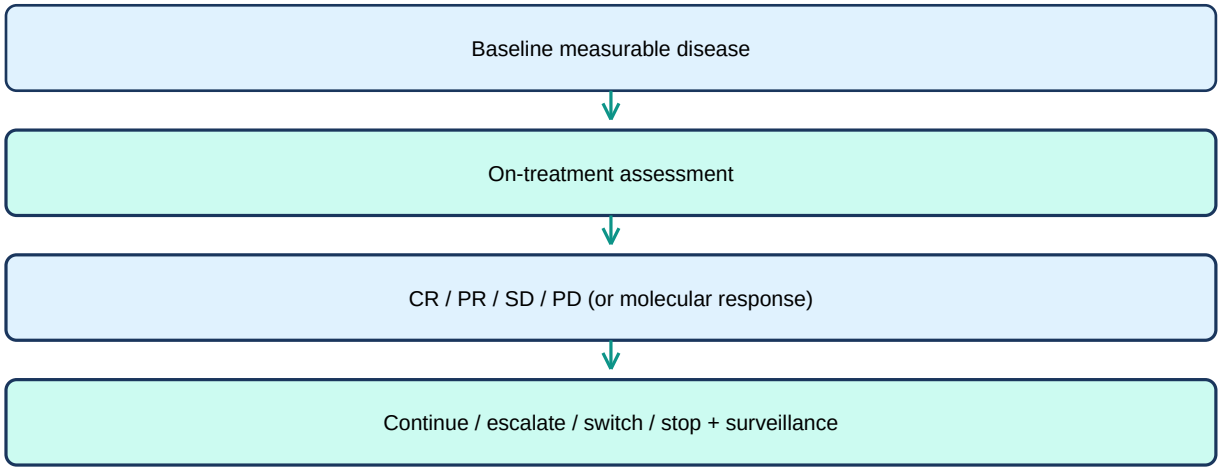


Figure: stepwise exam algorithm

13. Comparison tables useful for exams

Draw 2×N tables comparing old standard vs new standard; or drug A vs B (efficacy, toxicity, CNS activity, cost/access).

• Column ideas: target, dose, PFS, OS, grade ≥3 AEs, CNS activity, NCCN category.

• Another table: resistance mutation → preferred next drug.

Trial	Key result / exam takeaway
HERA / BCIRG-006 / NSABP B-31 / N9831	Adjuvant trastuzumab OS/DFS benefit
CLEOPATRA	Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MOSAIC / C-07 / IDEA	Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX
ATOMIC	FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred

Table: landmark clinical trials

14. Indian context, NCG & practical exam tips

National Cancer Grid (NCG) resource-stratified guidelines are frequently expected in Indian DM exams. Discuss biosimilars (trastuzumab, rituximab), generic TKIs, and center-based cell therapy availability.

• Document MDT decisions; clinical trial referral when standard options exhausted.

• Counsel financial toxicity transparently; prioritize highest-value Category 1 options.

• Infection prophylaxis algorithms crucial in hematologic topics (antifungal, antiviral, PJP).

15. Sample ‘examiner hot spots’ & model closing paragraph

Model closing: “In summary, management of this problem is biomarker-driven. Current standards are supported by phase III trials and reflected in NCCN 2026/2027 updates. Sequencing should maximize OS and QoL while mitigating class-specific toxicities. Enrollment in clinical trials and adherence to MDT pathways remain integral.”

Hot spots: exact doses, one HR/median, resistance mechanism, NCCN category, and a drawn algorithm.

EXAM PEARLS / HALLMARKS

- If stuck on memory: write mechanism → class → one trial → one dose → one toxicity → NCCN 2026 line.
- Never contradict yourself between algorithm and trial table.

A. Epidemiology & burden (write ½–1 page)

Begin with global and Indian epidemiology when known (GLOBOCAN / ICMR / NCRP). State age distribution, sex ratio, and risk factors (tobacco, infection-related cancers such as HPV/HBV/HCV/H. pylori/HIV, obesity, inherited syndromes).

Explain why incidence or mortality trends matter for screening and resource allocation. For rare diseases, state incidence per 100,000 and referral-center patterns.

Link epidemiology to prevention: vaccination (HPV, HBV), tobacco control, hepatitis treatment, aspirin in selected molecular CRC, and lifestyle (exercise post-adjuvant CRC per NCCN 2026).

- One sentence on screening relevance (mammography, LDCT, FIT/colonoscopy, low-dose CT criteria).
- Mention hereditary fraction if applicable (Lynch, HBOC, FAP, MEN, VHL, HPRCC).
- State stage distribution at diagnosis in LMICs vs HIC if relevant to treatment sequencing.

B. Historical evolution → modern standard (timeline)

Examiners like a short timeline: historical surgery/RT/chemo era → targeted therapy era → immunotherapy/ADC/cellular therapy era.

Highlight inflection trials that changed guidelines (e.g., imatinib IRIS; trastuzumab HERA; checkpoint inhibitors; ADCs; CAR-T approvals).

End timeline with **NCCN 2026/2027** preferred regimens so the answer feels contemporary for Paper IV.

Therapeutic eras timeline

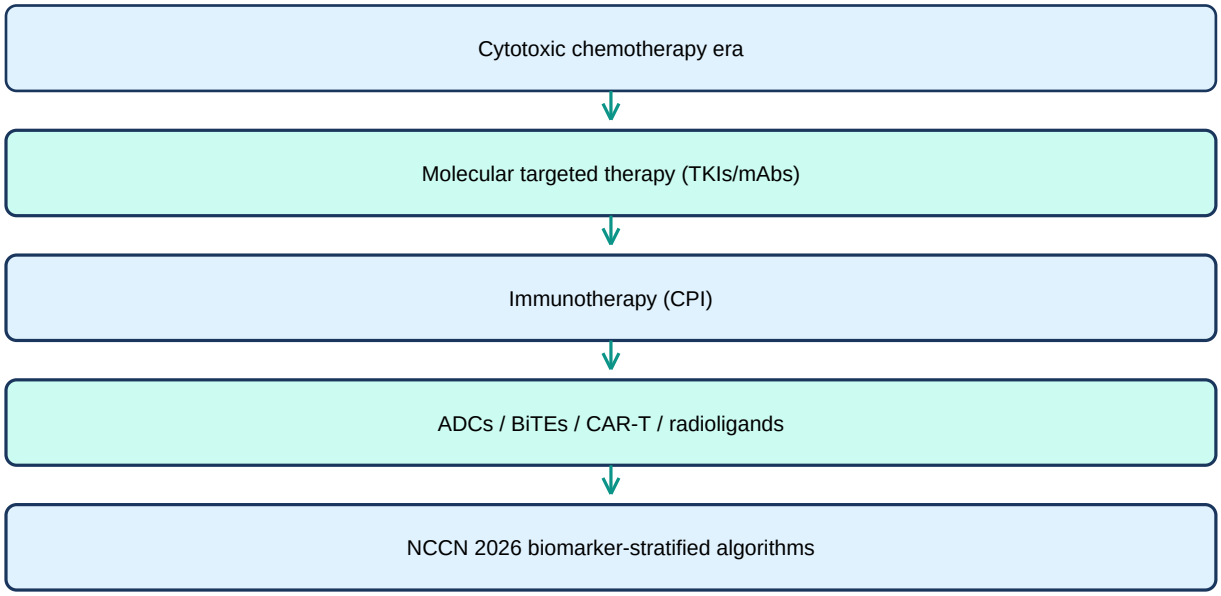


Figure: stepwise exam algorithm

C. Detailed differential diagnosis & pathology pearls

List 5–7 differentials and the IHC/molecular discriminators. Pathology reports should include grade, LVI, margins, nodes, and required biomarkers.

For hematologic topics: morphology → flow cytometry → cytogenetics → NGS panel → integrated WHO2022/ICC diagnosis.

For CUP/unknown primary style questions: IHC algorithm (CK7/CK20, TTF1, CDX2, GATA3, PAX8, synaptophysin) plus NGS and MSI/HER2.

- Always write ‘correlate clinically with radiology and tumor markers’.
- State pre-analytical caveats (decalcification, fixation time) affecting HER2/ER assays.
- Liquid biopsy complements but does not always replace tissue.

Morphology	Light microscopy pattern
IHC panel	Lineage + predictive markers
Cytogenetics	Karyotype / FISH fusions
NGS	SNVs, indels, CNAs, MSI, TMB
Integrated Dx	WHO/ICC + AJCC stage

Figure: mechanism / pathway map

D. Multidisciplinary care pathway & sequencing

Describe roles: medical oncology, surgical oncology, radiation oncology, radiology, pathology, palliative care, genetics, psycho-oncology, rehabilitation, and clinical trials unit.

Sequencing principles: treat drivers first when OS-proven; use neoadjuvant therapy to downstage and guide adjuvant escalation; avoid overlapping severe toxicities; protect future options (e.g., cumulative cardiotoxicity, neuropathy).

For oligometastatic disease discuss local therapy (surgery/SBRT/ablation) plus systemic therapy — prostate NCCN 2026 explicitly incorporates metastasis-directed therapy in selected APMS disease.

- Document MDT minutes in the file — also good exam phrase.
- Re-staging after neoadjuvant before definitive local therapy.
- Early palliative care improves QoL and sometimes survival in metastatic settings.

Clinical algorithm

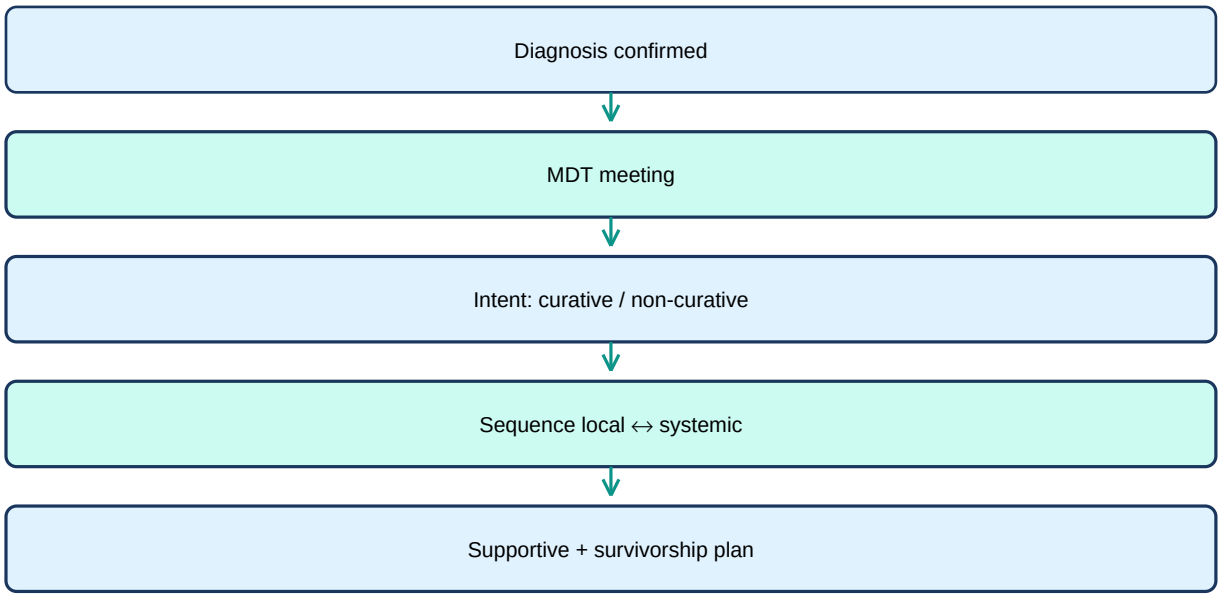


Figure: stepwise exam algorithm

E. Antibody–drug conjugate deep notes

ADC anatomy: mAb + linker + payload. Cleavable linkers + permeable payloads → bystander effect (T-DXd). Non-cleavable (T-DM1) more intracellularly restricted.

Payloads: microtubule inhibitors (DM1, MMAE) vs topoisomerase-I (DXd, SN-38).

NCCN 2026 expands ADCs across tumors: T-DXd, Dato-DXd, SG, EV, mirvetuximab, belantamab combos, loncastuximab.

mAb	Targets HER2/TROP2/Nectin-4/FRα/BCMA/CD30
Linker	Cleavable vs non-cleavable
Payload	Topo-I or microtubule toxin
Bystander	Kills neighboring low-antigen cells
Key toxicity	ILD / neuropathy / hyperglycemia / ocular

Figure: mechanism / pathway map

G. Supportive care mini-chapter

Antiemetics (olanzapine-containing regimens for HEC), G-CSF primary/secondary prophylaxis, antimicrobial prophylaxis, mucositis care, nutrition (including TPN indications/contraindications), pain ladder + interventional techniques, thrombosis prevention (Khorana), growth factors, transfusion thresholds.

Psychosocial distress screening, fertility preservation referral before gonadotoxic therapy, and survivorship clinics.

- Febrile neutropenia: risk assess; anti-pseudomonal beta-lactam; add antifungal if persistent.
- Checkpoint diarrhea: rule out infection then steroids promptly.
- ADC ILD algorithms — patient education on dyspnea.

H. Model answer skeleton (time plan for 35 minutes)

Minutes 0–3: definition + epidemiology + diagram outline. Minutes 3–12: biology + diagnosis. Minutes 12–25: management algorithm with doses/trials. Minutes 25–30: toxicities + NCCN 2026. Minutes 30–35: summary pearls + underline Category 1 regimens.

- Leave a half page blank early for a neat flowchart drawn once.
- Underline drug doses.
- Box the NCCN 2026 bullets.

EXAM PEARLS / HALLMARKS

- Currency + clarity > exhaustive rare regimens.
- Named trial + number beats unnamed claim.
- NCCN 2026 phrase is expected in Paper IV recent advances.

K. Pitfalls, contraindications & ‘do not miss’ warnings

List absolute/relative contraindications of key drugs (e.g., baseline ILD caution for T-DXd; severe autoimmune disease caution for CPI; T315I needs ponatinib/asciminib not 1G/2G TKI alone; ATRA differentiation syndrome in APL; TLS with venetoclax/Blincyto; HBV reactivation with anti-CD20/CPI).

Drug–drug interactions and QT prolongation traps are classic viva points — mention checking concomitant meds (azoles, macrolides, acid suppressants with selected TKIs).

Imaging pitfalls: pseudo-progression on IO; flare on bone scan; PET Deauville interpretation after immunomodulatory therapy.

- Always screen hepatitis B before anti-CD20, anthracyclines, and many systemic regimens.
- Pregnancy test before teratogenic agents (IMiDs — REMS; MTX; TKI counseling).
- Do not combine overlapping QT-prolonging agents without ECG plan.
- Document performance status — unfit vs fit changes entire AML/lymphoma pathway.

Clinical algorithm

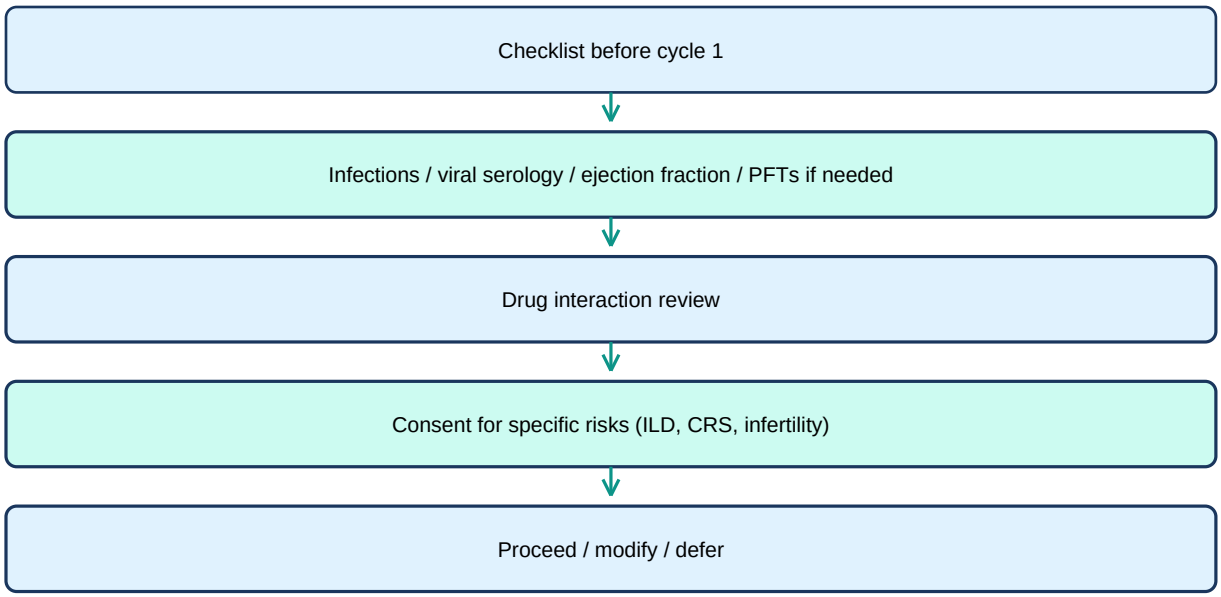


Figure: stepwise exam algorithm

L. Case vignette practice (apply the algorithm)

Invent a short case in the answer: age, stage, biomarkers, comorbidities → your exact regimen with dose → expected monitoring → what you will do at progression. This shows applied knowledge and often gains examiner goodwill.

Example template: “45-year-old, ECOG 1, biomarker X positive, no contraindication Y → start regimen Z at dose… per trial T / NCCN 2026 preferred. Assess response at interval N. At progression, re-profile for resistance mutation R and switch to agent S.”

Add one geriatric or organ-dysfunction variant to show nuance (dose reduce, alternate preferred regimen, or best supportive care).

- Vignette 1: fit, biomarker-positive — preferred Cat 1 therapy.
- Vignette 2: comorbid/unfit — alternative regimen.
- Vignette 3: progression with resistance mutation — sequenced next line.
- Vignette 4: severe toxicity — hold, treat, rechallenge decision.

M. One-page ‘wall chart’ summary (draw/box this)

Reserve the penultimate page for a wall-chart style summary with four quadrants: (1) Biology (2) Work-up (3) Treatment lines (4) NCCN 2026 bullets. This mirrors how many toppers present long answers and helps hit the page target with high signal density.

Quadrant 1	Hallmark gene / pathway / figure
Quadrant 2	Must-order tests + staging
Quadrant 3	1L / 2L / 3L with doses
Quadrant 4	Trials + NCCN 2026 + toxicities
Center	MDT + patient goals

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

- Wall chart + flowchart + trial table = visual completeness.
- Write ‘NCCN 2026’ in quadrant 4 explicitly.

I2. Full-length essay addendum — tables to copy into the answer book

For 15-mark elaborations, add two hand-drawn tables: (Table A) Drug–dose–pearl; (Table B) Trial–population–endpoint–practice change. Then write a short paragraph on quality-of-life and patient-reported outcomes, which modern trials increasingly report.

Discuss biomarker evolution on therapy (liquid biopsy cadence), and when to stop therapy (duration of CPI, TFR in CML, maintenance stopping rules).

Include a brief note on second malignancies and late effects when curative-intent therapy is discussed (alkylators, RT fields, PARPi MDS/AML risk, CHL survivors).

- Table A columns: Drug | Dose | Key toxicity | Pearl
- Table B columns: Trial | Arm | Primary endpoint | Result | Change in practice
- One paragraph on shared decision-making and financial counseling

Trial	Key result / exam takeaway

HERA / BCIRG-006 / NSABP B-31 / N9831	Adjuvant trastuzumab OS/DFS benefit
CLEOPATRA	Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MOSAIC / C-07 / IDEA	Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX
ATOMIC	FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred
ZUMA-1 / JULIET / TRANSCEND	Axi-cel; tisa-cel; liso-cel LBCL
ZUMA-7 / TRANSFORM / BELINDA	2L CAR-T vs SOC

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:*

• Expanding ADC footprint across NCCN 2026: T-DXd (HER2 breast/gastric/NSCLC/urothelial), Dato-DXd (EGFR+ NSCLC), sacituzumab govitecan (breast/urothelial), mirvetuximab (ovarian FRα), enfortumab vedotin (urothelial), belantamab combinations (myeloma), loncastuximab (B-cell).

I. Extended notes — resistance, sequencing & future directions

Resistance biology should occupy a dedicated half-page: on-target secondary mutations, bypass signaling, lineage plasticity (e.g., EGFR NSCLC → SCLC), antigen loss after cellular therapy, and immunosuppressive TME changes after CPI.

Sequencing principles: use the agent with largest OS benefit early when toxicities allow; reserve cross-resistant drugs; consider cumulative neuropathy/cardiotoxicity; plan CNS-active agents if brain metastases are present or likely.

Future directions appropriate for 2026 answers: tumor-agnostic indications (NTRK, RET, BRAF, MSI-H, TMB-H, HER2-mut), radioligand therapy, CAR-NK/off-the-shelf products, ctDNA-guided escalation/de-escalation, AI decision-support, and microbiome modulation adjuncts.

Health services angle: real-world evidence validating trial results in older/comorbid patients; digital remote toxicity monitoring; and equity of molecular testing access — themes also reflected in recent Paper IV questions.

- Write one ‘if progression then…’ resistance table.
- Mention at least one tumor-agnostic approval if biology overlaps.
- Close with trial enrollment as a standard option.

J. Quick-revision flash cards (last 1–2 pages)

Convert the topic into 8–10 flash cards mentally: (1) definition (2) hallmark gene (3) key test (4) 1L preferred (5) dose (6) trial (7) top toxicity (8) NCCN 2026 update (9) Indian/NCG pearl (10) one differential.

- Flash 1: One-line definition
- Flash 2: Driver / pathway
- Flash 3: Mandatory biomarker

- Flash 4: NCCN preferred 1L
- Flash 5: Exact dose
- Flash 6: Landmark phase III + endpoint
- Flash 7: Grade ≥3 toxicity to warn
- Flash 8: 2026 guideline change
- Flash 9: Resistance next step
- Flash 10: Supportive care must-do

EXAM PEARLS / HALLMARKS

- If time-poor in the hall: draw algorithm + trial table + NCCN box — captures most marks.
- Keep handwriting large for doses and HRs.

Final box — paste into last page of answer sheet

Re-list NCCN 2026 updates here so the examiner sees currency even if earlier pages are rushed.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:*
- Expanding ADC footprint across NCCN 2026: T-DXd (HER2 breast/gastric/NSCLC/urothelial), Dato-DXd (EGFR+ NSCLC), sacituzumab govitecan (breast/urothelial), mirvetuximab (ovarian FRα), enfortumab vedotin (urothelial), belantamab combinations (myeloma), loncastuximab (B-cell).

Final NCCN 2026 box

Copy these bullets onto the last page of your answer booklet for Paper IV currency marks.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:*
- Expanding ADC footprint across NCCN 2026: T-DXd (HER2 breast/gastric/NSCLC/urothelial), Dato-DXd (EGFR+ NSCLC), sacituzumab govitecan (breast/urothelial), mirvetuximab (ovarian FRα), enfortumab vedotin (urothelial), belantamab combinations (myeloma), loncastuximab (B-cell).

Disclaimer: Compiled for DM exam revision from published trials and NCCN 2026/2027 public summaries (ASCO Post, OncLive, JNCCN selections, NCCN category list). Always verify the current full NCCN PDF, FDA/CDSCO labels, and institutional protocols before patient care. Doses require BSA/organ-function adjustment.